GSC-X Cardiac Sandbox

Cardiac Validation V1:
Phenotype-Aware Early Warning, Metastability Monitoring and Virtual Pacemaker Guard Architecture

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PV 2026-216 (GSC-F)
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Abstract

The GSC-X Cardiac Sandbox investigates whether executive-state architectures can provide meaningful early-warning, metastability monitoring and phenotype-aware decision support for cardiac rhythm dynamics.

This work presents Cardiac Validation V1, a research validation campaign based on public MIT-BIH Arrhythmia records and internal GSC-X cardiac benchmark outputs.

Instead, this study evaluates whether Context Memory (CM), Early Limit Indicator (ELI), Predictive Transition Memory (PTM), Metastability Reserve (MR), Metastability Risk Index (MRI) and phenotype-aware routing can provide a structured research framework for identifying fragile cardiac regimes with a median predictive lead of 23.75 beats before critical rhythm transitions occur.

Initial global CM+ELI benchmarks demonstrated improved event detection but also revealed an important limitation: a single global alarm threshold is not sufficient for reliable cardiac decision-making.

False-positive audits showed that increased sensitivity can produce excessive alarm burden.

Subsequent experiments therefore shifted toward phenotype-aware routing, collapse discrimination, cardiac reserve estimation and virtual pacemaker guard logic.

The strongest architectural result of Cardiac Validation V1 is the emergence of a phenotype-aware executive framework.

Records were not treated as identical signal streams.

Instead, cardiac behavior was interpreted through operating phenotypes, metastability reserve and executive state transitions.

In the broader Universal Metastability Kernel V1.1 validation campaign, the Cardiac branch was successfully integrated as the ninth validation domain.

Using the same shared Recovery–Stress–Persistence kernel and the same executive-state grammar, cardiac outputs were separated into HIGH_RESERVE, STABLE, FRAGILE, LOW_RESERVE and CRITICAL states.

This result supports the interpretation that cardiac dynamics can be represented as a metastability-reserve problem rather than as a simple threshold-alarm problem.

The proposed Virtual Pacemaker Guard concept is presented as a research architecture only.

It is not a medical device, not a clinical decision system and not safety-certified.

Nevertheless, the validation results suggest that phenotype-aware executive-state monitoring may represent a promising research direction for future low-burden, reserve-aware cardiac control systems.

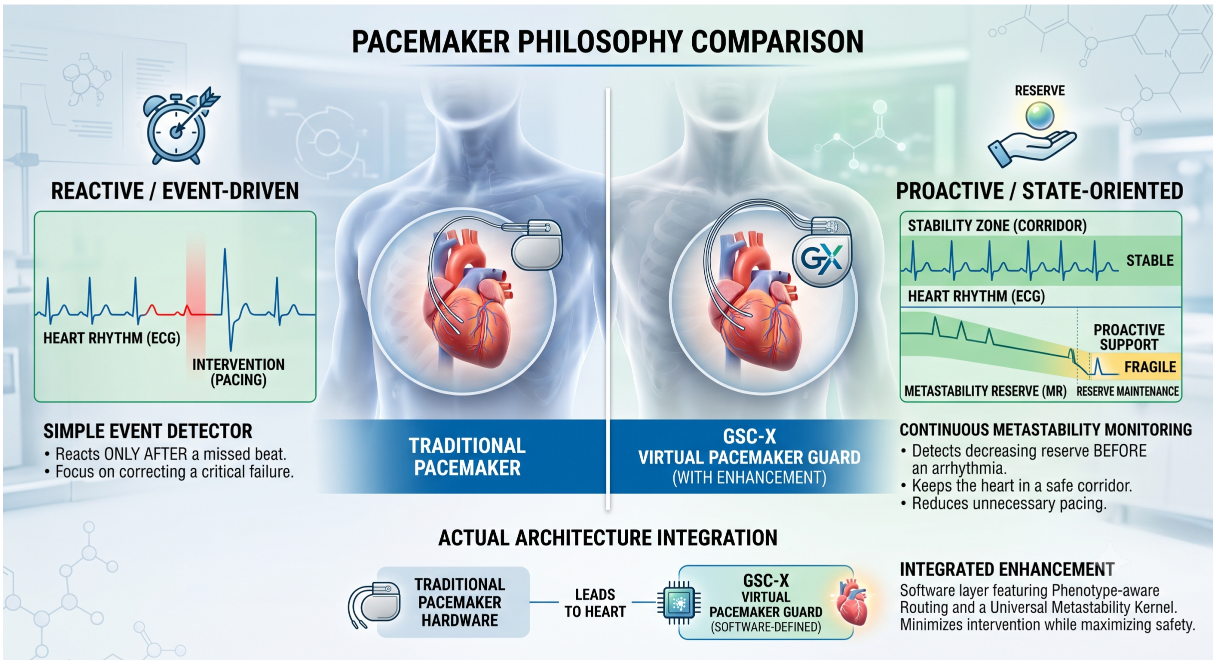

Figure 1: Conceptual overview of the proposed GSC-X Virtual Pacemaker Guard. In contrast to conventional event-driven pacemakers, which primarily react after predefined pacing criteria are met, the proposed architecture continuously estimates cardiac metastability reserve using the Recovery–Stress–Persistence framework and the Universal Metastability Kernel. Executive decisions are guided by phenotype-aware routing and reserve preservation, enabling proactive state-oriented guidance rather than purely reactive pacing. The architecture is presented as a research concept intended for computational validation and future investigation rather than clinical application.

Cardiac Validation Summary
Dataset: MIT-BIH Arrhythmia Database
Validation stages: 8
Executive generations: CM ELI PTM COLI APB Phenotype Router Virtual Pacemaker Guard Universal Kernel
Integrated as: 9th Universal Metastability Kernel Domain

1 Introduction

Cardiac rhythm disorders represent one of the most challenging problems in physiological monitoring because the transition from stable electrical activity to pathological rhythm often develops progressively rather than instantaneously.

Most conventional monitoring systems rely on predefined thresholds or single-event detectors that trigger an alarm only after a measured variable exceeds a fixed limit. Although this strategy is effective for many well-defined abnormalities, it frequently produces two fundamental problems. First, excessive sensitivity generates a large number of false-positive alarms. Second, conservative thresholds may delay detection until the physiological state has already deteriorated.

These limitations suggest that cardiac monitoring should not be viewed solely as an event-detection problem. Instead, the underlying physiological system may be interpreted as a continuously evolving dynamical process whose internal stability changes over time.

The GSC-X architecture approaches this problem from the perspective of metastability. Rather than asking whether an abnormal beat has already occurred, the Executive Core continuously evaluates the balance between recovery mechanisms, accumulated physiological stress and the persistence of destabilizing processes.

Within this framework, every cardiac recording is interpreted as a dynamic trajectory through an executive state space instead of a sequence of isolated arrhythmia events.

The present work introduces the GSC-X Cardiac Sandbox, a research environment designed to evaluate executive-state monitoring using publicly available cardiac databases together with the common GSC-X Executive Core.

Unlike classical classification studies that focus primarily on diagnostic accuracy, the objective of the Cardiac Sandbox is to investigate whether higher-level executive quantities can describe the global condition of a cardiac system.

The Executive Core operates using a common set of metastability variables:

- Recovery
- Stress
- Persistence

These quantities are transformed into a universal executive representation consisting of:

- Recovery Stability Index (RSI)
- Metastability Stress Index (MSI)
- Reserve Stability Balance (RSB)
- Metastability Reserve (MR)
- Metastability Risk Index (MRI)
- Executive State

- Executive Transition
- Executive Decision

One of the central hypotheses investigated in this work is that the same executive architecture can be applied across heterogeneous dynamical systems.

During the broader Universal Metastability Kernel V1.1 validation campaign, identical executive equations were successfully reused across Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear validation environments.

The Cardiac Sandbox represents the biological validation branch of this unified framework.

Instead of developing a cardiac-specific mathematical model, cardiac telemetry is projected onto the same Recovery–Stress–Persistence representation used throughout the remaining GSC-X validation domains.

Consequently, cardiac dynamics become directly comparable with industrial processes, autonomous systems, energy networks and other complex dynamical environments using one common executive interpretation layer.

2 GSC-X Cardiac Executive Architecture

The Cardiac Sandbox is built upon the common GSC-X Executive Core rather than a cardiac-specific decision algorithm.

The Executive Core separates domain-dependent telemetry processing from domain-independent executive reasoning.

Consequently, only the telemetry layer is specific to cardiac signals, while all higher layers remain identical to those validated in Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear environments.

The Executive processing pipeline is illustrated conceptually as

$$
\begin{gathered}
\text{Telemetry}\\
\downarrow\\
\text{Context Memory (CM)}\\
\downarrow\\
\text{Early Limit Indicator (ELI)}\\
\downarrow\\
\text{Predictive Transition Memory (PTM)}\\
\downarrow\\
\text{Recovery} - \text{Stress} - \text{Persistence}\\
\downarrow\\
\text{RSI} - \text{MSI} - \text{RSB}\\
\downarrow\\
\text{MR} - \text{MRI}\\
\downarrow\\
\text{Executive State}\\
\downarrow\\
\text{Executive Transition}\\
\downarrow\\
\text{Executive Decision}
\end{gathered}
$$

The lower part of the architecture (telemetry acquisition and feature extraction) depends on the monitored domain.

In the Cardiac Sandbox the telemetry originates from ECG recordings and beat annotations.

In contrast, every layer beginning with Recovery–Stress–Persistence is identical across all validated GSC-X domains.

This separation enables a common executive representation independent of the physical nature of the monitored system.

Recovery estimates the capability of the system to maintain organized behaviour.

Stress represents destabilizing influences acting on the system.

Persistence quantifies how long destabilizing behaviour survives despite recovery mechanisms.

Together these three quantities define the internal metastability state of the monitored process.

The Executive Core derives five executive indices from these variables.

- Recovery Stability Index (RSI)
- Metastability Stress Index (MSI)
- Reserve Stability Balance (RSB)
- Metastability Reserve (MR)
- Metastability Risk Index (MRI)

These indices provide a continuous representation of the system condition rather than a binary alarm signal.

The final Executive Layer converts the continuous executive indices into discrete operational states.

Five executive states are currently defined:

- HIGH_RESERVE
- STABLE
- FRAGILE
- LOW_RESERVE
- CRITICAL

Unlike traditional threshold-based monitoring, executive states describe the current operating regime of the system rather than merely indicating whether a predefined alarm limit has been exceeded.

This representation allows the Executive Core to reason about gradual degradation, reserve exhaustion and transition dynamics before complete instability develops.

Within the Cardiac Sandbox these executive states form the basis of the proposed Virtual Pacemaker Guard architecture, whose objective is to minimise unnecessary interventions while maintaining sufficient physiological reserve.

The Executive Core therefore acts as a common decision layer capable of transforming heterogeneous telemetry into a unified metastability-oriented description of system behaviour.

3 Validation Dataset and Experimental Campaign

The Cardiac Sandbox was evaluated using publicly available electrocardiographic recordings from the MIT-BIH Arrhythmia Database.

Rather than evaluating a single binary classifier, the objective of the validation campaign was to investigate the complete executive-state architecture under progressively more realistic operating conditions.

The experimental workflow evolved through multiple validation stages, where the outcome of each stage motivated the development of the following generation.

The validation campaign therefore represents an iterative refinement of the Executive Core rather than a single benchmark experiment.

3.1 Dataset

The primary validation dataset consisted of ten representative MIT-BIH Arrhythmia records:

100–109

These recordings contain approximately

- 650 000 ECG samples,
- two ECG channels,
- sampling frequency of 360 Hz,
- manually annotated cardiac events.

The analysed annotations included normal and pathological cardiac rhythms together with ventricular, atrial and conduction abnormalities.

The purpose of the dataset was not diagnostic classification itself but validation of executive-state monitoring under heterogeneous physiological behaviour.

3.2 Validation Strategy

Rather than evaluating a single detector, the Cardiac Sandbox was developed through a sequence of complementary validation campaigns, each targeting a different aspect of the Executive Core architecture.

The complete validation strategy consisted of the following stages:

1. Event Audit
2. Context Memory (CM) + Early Limit Indicator (ELI) Executive Validation
3. False Positive Audit
4. Breathing Executive Layer
5. Collapse Executive Layer
6. COLI Executive Layer
7. APB-aware Executive Layer
8. Phenotype-aware Routing
9. Universal Metastability Kernel Integration

Each validation stage introduced additional executive capabilities while preserving compatibility with the common GSC-X Executive Core architecture. Rather than replacing previous generations, successive stages progressively refined the interpretation of cardiac dynamics, ultimately culminating in phenotype-aware executive routing and cross-domain integration within the Universal Metastability Kernel V1.1.

3.3 Event Audit

The first validation stage analysed the complete annotation stream and established the baseline distribution of cardiac events.

A total of 3216 annotated events were processed.

The dominant rhythm disturbances were left bundle branch block (L), ventricular ectopic beats (V) and atrial premature beats (A).

This audit provided the reference dataset for all subsequent executive evaluations.

3.4 False Positive Audit

The second stage investigated the practical behaviour of CM+ELI early-warning logic.

Although the detector substantially increased the number of correctly detected events, the accompanying increase in false-positive alarms demonstrated that a single global alarm threshold is insufficient for practical executive decision making.

This result became one of the central motivations for phenotype-aware executive routing.

3.5 Executive Layer Development

Subsequent validation stages progressively incorporated additional executive mechanisms.

Breathing ELI investigated respiratory-related temporal dynamics.

Collapse Discriminator focused on identifying rapid transitions toward unstable cardiac regimes.

COLI and APB-aware executive layers evaluated alternative strategies for balancing detection sensitivity, lead time and false-positive burden.

These experiments demonstrated that no single global detector simultaneously optimizes all performance metrics.

Instead, different physiological regimes require different executive behaviour.

3.6 Phenotype-aware Routing

One of the principal findings of the Cardiac Sandbox was that individual recordings naturally cluster into distinct operating phenotypes.

Instead of applying one universal detector to every patient, executive routing selects a monitoring strategy according to the observed metastability characteristics.

The resulting phenotype-aware Executive Router significantly reduced false-positive burden while maintaining useful predictive lead times.

This architecture represents the foundation of the proposed Virtual Pacemaker Guard introduced later in this work.

3.7 Universal Kernel Validation

The final validation stage projected the Cardiac Sandbox into the Universal Metastability Kernel V1.1.

Recovery, Stress and Persistence variables derived from cardiac telemetry were processed using exactly the same executive equations previously validated in Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear environments.

Consequently, cardiac dynamics became directly comparable with other complex engineering systems while preserving a single executive interpretation framework.

The successful integration of Cardiac as the ninth validation domain demonstrates the portability of the proposed Executive Core across fundamentally different dynamical systems.

4 Experimental Results

The Cardiac Sandbox validation campaign demonstrated that executive-state monitoring evolves progressively as additional metastability information is incorporated into the decision process.

Rather than relying on a single detector, the validation campaign evaluated multiple generations of the Executive Core under identical cardiac telemetry.

The principal objective was to investigate the balance between early detection, false-positive burden and executive-state stability.

4.1 Event Detection

The initial CM+ELI experiments demonstrated that executive telemetry substantially improves the detection of several cardiac event classes.

Particularly noticeable improvements were observed for atrial premature beats and ventricular abnormalities compared with the baseline detector.

However, the increased sensitivity also produced a significant increase in false-positive alarms.

This behaviour confirmed that maximizing sensitivity alone does not necessarily improve executive decision quality.

Consequently, additional executive layers were required.

4.2 False Positive Behaviour

A dedicated false-positive audit revealed one of the central findings of the Cardiac Sandbox.

Although CM+ELI successfully detected substantially more cardiac events, excessive alarm generation became the dominant practical limitation.

This observation motivated the transition from a purely threshold-based detector toward phenotype-aware executive monitoring.

The results indicate that executive quality should not be evaluated solely by detection rate.

Instead, detection quality, lead time and intervention burden must be considered simultaneously.

4.3 Phenotype-aware Executive Routing

The most significant improvement was achieved after introducing phenotype-aware routing.

Instead of assuming identical behaviour for every cardiac recording, the Executive Router classified recordings into distinct operating phenotypes and selected an appropriate executive strategy.

The final Executive Router V3 achieved

- mean false-positive rate: 36.30 FP/hour (all records),
- active-record precision: 0.7397,
- active-record detection: 0.4809,
- median predictive lead: 23.75 beats.

These results demonstrate that phenotype-aware routing substantially improves executive stability compared with a globally tuned detector.

The experiments therefore support the hypothesis that cardiac executive monitoring should adapt to the physiological operating regime rather than relying on one universal threshold.

4.4 Executive State Representation

Unlike conventional arrhythmia detectors, the Executive Core continuously estimates the metastability reserve of the monitored cardiac system.

Instead of producing a binary alarm, the Executive Core represents cardiac behaviour through five executive operating states:

$$\begin{array}{c} \text{HIGH\_RESERVE} \\ \downarrow \\ \text{STABLE} \\ \downarrow \\ \text{FRAGILE} \\ \downarrow \\ \text{LOW\_RESERVE} \\ \downarrow \\ \text{CRITICAL} \end{array}$$

This representation enables gradual tracking of physiological degradation before complete instability develops.

The executive-state representation therefore provides substantially richer information than a simple event detector.

4.5 Universal Metastability Kernel

The final validation stage projected all cardiac executive outputs into the Universal Metastability Kernel V1.1.

The same Recovery–Stress–Persistence equations previously validated across engineering domains were applied without modification.

Cardiac became the ninth integrated validation domain.

The resulting executive-state distribution was

Executive State	Units	Mean MRI
HIGH_RESERVE	55	0.0605
STABLE	5	0.2623
FRAGILE	1	0.4464
LOW_RESERVE	9	0.7505
CRITICAL	24	0.9653

Table 1: Cardiac Universal Metastability Kernel V1.1 Results

Executive State	Units	Mean MR	Mean MRI	Mean RSB
HIGH_RESERVE	55	0.939478	0.060522	0.693664
STABLE	5	0.737737	0.262263	0.322682
FRAGILE	1	0.553636	0.446364	0.185895
LOW_RESERVE	9	0.249517	0.750483	-0.034887
CRITICAL	24	0.034651	0.965349	-0.522228

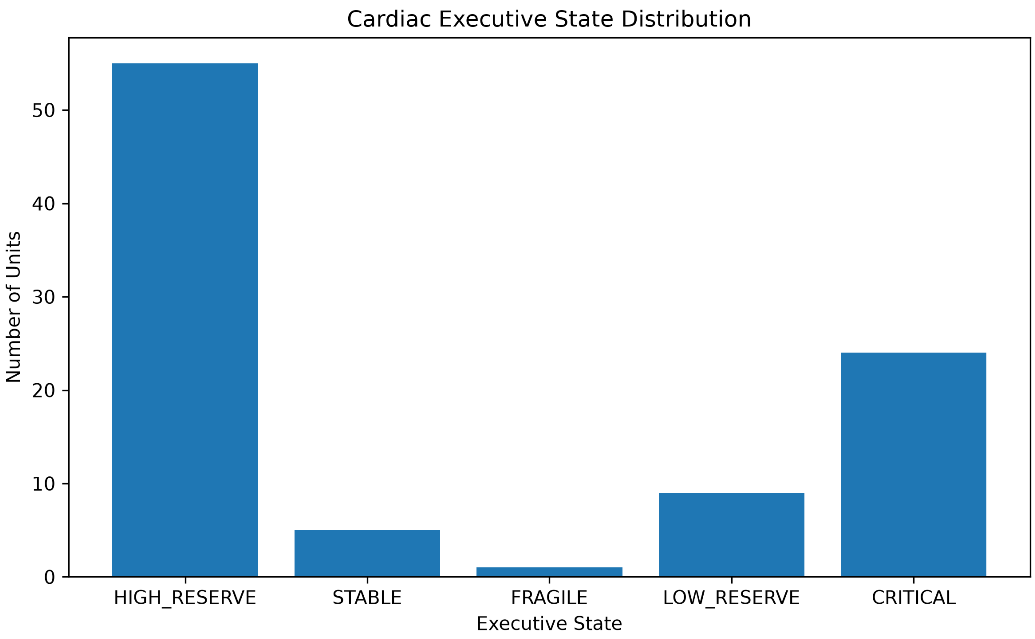

Figure 2: Distribution of Cardiac Executive States generated by the Universal Metastability Kernel V1.1. The cardiac branch spans the complete executive-state spectrum from HIGH_RESERVE to CRITICAL, indicating that the kernel provides graded metastability interpretation rather than a binary alarm output.

Importantly, cardiac recordings did not collapse into a single executive category.

Instead, the Universal Kernel produced a continuous spectrum of metastability states ranging from highly stable cardiac behaviour to severely degraded operating regimes.

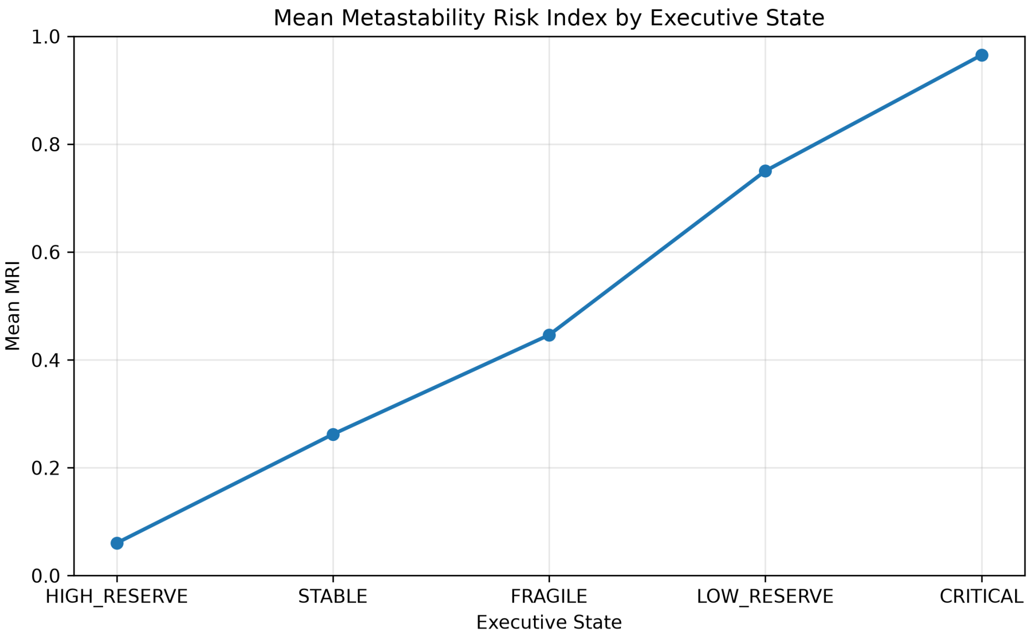

Figure 3: Mean Metastability Risk Index (MRI) by Cardiac Executive State. MRI increases monotonically from HIGH_RESERVE to CRITICAL, supporting the interpretation that the executive-state grammar represents progressive loss of cardiac metastability reserve.

The monotonic ordering represents an internal consistency check of the Universal Metastability Kernel.

This result demonstrates that cardiac telemetry can be interpreted using the same executive representation employed across batteries, industrial systems, energy networks, plasma control, robotics and nuclear simulations.

The successful integration of cardiac physiology into the common executive framework represents one of the strongest cross-domain validation results obtained during the development of **GSC-X Executive Core V1.1.**

5 Virtual Pacemaker Guard Architecture

One of the principal outcomes of the Cardiac Sandbox is that executive-state monitoring naturally leads to a different pacing philosophy than conventional threshold-based control.

Traditional pacemakers primarily react after predefined electrical criteria have already been satisfied.

Although such systems are highly effective for many established clinical indications, their decision process is generally event-driven.

In contrast, the proposed GSC-X Virtual Pacemaker Guard treats cardiac rhythm as a continuously evolving dynamical system whose metastability reserve changes over time.

Rather than asking whether a single abnormal beat has occurred, the Executive Core continuously estimates the internal operating condition of the cardiac system.

The proposed research architecture consists of the following executive pipeline

$$\begin{array}{c}\text{ECG Telemetry}\\ \downarrow\\ \text{Feature Extraction}\\ \downarrow\\ \text{Phenotype Classification}\\ \downarrow\\ \text{Context Memory (CM)}\\ \downarrow\\ \text{Early Limit Indicator (ELI)}\\ \downarrow\\ \text{Predictive Transition Memory (PTM)}\\ \downarrow\\ \text{Recovery} - \text{Stress} - \text{Persistence}\\ \downarrow\\ \text{RSI} - \text{MSI} - \text{RSB}\\ \downarrow\\ \text{MR} - \text{MRI}\\ \downarrow\\ \text{Executive State}\\ \downarrow\\ \text{Executive Decision}\\ \downarrow\\ \text{Virtual Pacemaker Guard}\end{array}$$

Within this framework the pacing logic is not driven by a single threshold crossing.

Instead, intervention is interpreted as an executive decision based on the overall metastability state of the monitored cardiac system.

The Executive Core continuously estimates the available physiological reserve through Metastability Reserve (MR) and Metastability Risk Index (MRI).

A gradual decrease in MR together with persistent stress accumulation may indicate progressive degradation of cardiac stability before catastrophic rhythm transitions occur.

Consequently, executive decisions become state-oriented rather than event-oriented.

Instead of binary pacing recommendations, the Executive Core may distinguish multiple operational regimes ranging from simple observation to protective intervention.

The executive-state interpretation therefore provides considerably richer information than a conventional alarm generator.

An additional consequence of the phenotype-aware architecture is the reduction of unnecessary interventions.

Different cardiac phenotypes exhibit substantially different metastability dynamics.

Applying identical intervention logic to all patients therefore represents an unnecessary simplification.

The proposed Virtual Pacemaker Guard adapts executive reasoning to the observed operating phenotype while preserving the same executive equations.

The long-term objective is not to maximise intervention frequency.

The objective is not to increase pacing frequency, but to preserve physiological reserve while minimizing unnecessary interventions..

Conceptually, the Virtual Pacemaker Guard therefore follows three principles:

1. continuous estimation of physiological reserve,
2. intervention only during persistent reserve degradation,
3. verification that recovery actually increases after intervention.

This approach differs fundamentally from conventional threshold-triggered control because intervention becomes only one possible executive action rather than the primary objective of the monitoring system.

The architecture presented in this work should be regarded as a research concept intended for simulation studies and algorithmic validation.

It is not intended for clinical use and has not undergone regulatory evaluation.

6 Discussion

The Cardiac Sandbox represents the biological validation branch of the GSC-X Executive Core.

Table 2: Universal Metastability Kernel V1.1 Cross-Domain Integration

Domain	Integrated	Role in Validation
Battery	Yes	Degradation and end-of-life dynamics
NASA CMAPSS	Yes	Turbofan-engine prognostics
OpenCARP	Yes	Cardiac reentry corridor simulation
Fusion	Yes	Plasma instability and persistence
Industrial	Yes	Process stability and hard-delay control
Energy	Yes	Grid resilience and cascade risk
Robotics	Yes	Autonomous control and task instability
Nuclear	Yes	Operating-map and transmutation regimes
Cardiac	Yes	Biological rhythm metastability

Unlike the remaining validation environments, which originate from engineering systems, the cardiac domain introduces complex physiological dynamics characterized by substantial inter-patient variability, nonlinear behaviour and heterogeneous pathological mechanisms.

Despite these differences, the same Executive Core was successfully applied without modifying its higher-level architecture.

The validation campaign demonstrated that cardiac telemetry can be transformed into a common executive representation using the Recovery–Stress–Persistence framework.

Consequently, executive quantities such as RSI, MSI, RSB, MR and MRI become independent of the original telemetry source and provide a unified description of system behaviour.

One of the most important observations obtained during the validation campaign was that a single global decision threshold is insufficient for heterogeneous cardiac recordings.

Initial CM+ELI experiments improved event detection but simultaneously increased false-positive alarms.

Subsequent validation stages demonstrated that executive performance improved when recordings were interpreted through phenotype-aware routing rather than through globally tuned thresholds.

This observation is consistent with the intuitive expectation that different physiological operating regimes require different executive strategies.

The Executive Core therefore shifts the focus from isolated event detection toward continuous estimation of metastability reserve.

Within this framework, abnormal beats are interpreted as one manifestation of a broader dynamical process rather than independent events.

The resulting executive states provide information not only about the current condition but also about the direction of system evolution.

An additional outcome of this work is the successful integration of Cardiac into the Universal Metastability Kernel V1.1.

The same executive equations previously validated in Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear environments were reused without modification.

Cardiac therefore became the biological validation branch of the common executive architecture.

This cross-domain validation supports the hypothesis that executive-state reasoning may represent a domain-independent abstraction layer applicable to heterogeneous dynamical systems.

However, several limitations should be acknowledged.

The present work was performed using publicly available offline datasets and retrospective validation procedures.

No prospective clinical evaluation has been performed.

The Executive Core was evaluated as a research framework rather than as a medical device.

Accordingly, no conclusions regarding clinical efficacy, patient outcomes or therapeutic benefit should be drawn from the present results.

Future work should therefore focus on larger cardiac datasets, prospective validation protocols, robustness against noisy physiological recordings, and simulation-based evaluation of Virtual Pacemaker Guard strategies under diverse cardiac conditions.

Nevertheless, the current validation campaign demonstrates that executive-state monitoring provides a structured and reproducible research framework for studying cardiac metastability using the same computational architecture already validated across multiple engineering domains.

7 Conclusion

This work presented the GSC-X Cardiac Sandbox as the biological validation branch of the GSC-X Executive Core.

Unlike conventional arrhythmia studies that primarily focus on event classification, the objective of this work was to investigate whether cardiac dynamics can be interpreted through a common executive-state architecture based on metastability monitoring.

The validation campaign demonstrated that cardiac telemetry can be transformed into a common executive representation using the Recovery–Stress–Persistence framework.

From these variables, the Executive Core derives a unified set of executive quantities including the Recovery Stability Index (RSI), Metastability Stress Index (MSI), Reserve Stability Balance (RSB), Metastability Reserve (MR) and Metastability Risk Index (MRI).

These executive quantities enable continuous estimation of system condition instead of binary threshold-triggered alarms.

A major finding of the validation campaign was that a single global detector is insufficient for heterogeneous cardiac recordings.

Although early CM+ELI experiments improved event detection, they also increased false-positive alarms.

Subsequent phenotype-aware routing demonstrated that executive monitoring benefits from adapting the decision strategy to the observed operating phenotype rather than applying identical thresholds to all recordings.

The final Cardiac Sandbox successfully integrated phenotype-aware routing, executive-state monitoring and the Universal Metastability Kernel V1.1.

Cardiac became the ninth validation domain using the same Executive Core previously validated across Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear environments.

Importantly, the executive equations remained unchanged throughout this cross-domain validation campaign.

The Universal Kernel produced a graded distribution of cardiac executive states ranging from HIGH_RESERVE through STABLE, FRAGILE and LOW_RESERVE to CRITICAL.

This result demonstrates that the Executive Core provides a continuous representation of physiological operating regimes instead of a binary healthy/failure classification.

The proposed Virtual Pacemaker Guard should be interpreted as a research architecture that illustrates how executive-state monitoring could be combined with phenotype-aware decision making in future simulation studies.

The present work does not propose a clinically validated pacing algorithm and should not be interpreted as evidence of clinical effectiveness.

Overall, the Cardiac Sandbox demonstrates that executive-state monitoring can provide a reproducible and interpretable research framework for studying cardiac dynamics while remaining fully compatible with the common GSC-X Executive architecture.

The results further support the broader hypothesis investigated throughout the GSC-X validation program: that a shared executive-state representation can be applied across heterogeneous complex dynamical systems without modifying the underlying executive kernel. Rather than proposing another arrhythmia detector, the Cardiac Sandbox demonstrates that cardiac dynamics can be interpreted as an evolving metastability process represented through a common executive-state architecture shared across heterogeneous dynamical systems.

8 Future Work

The current Cardiac Sandbox establishes the first generation of executive-state cardiac validation within the GSC-X framework.

Future development will focus on several research directions.

First, the phenotype-aware Executive Router will be extended to larger and more heterogeneous cardiac datasets in order to evaluate robustness across broader physiological variability.

Second, the Virtual Pacemaker Guard will be expanded with executive transition monitoring, allowing intervention strategies to depend not only on the current executive state but also on the observed direction of state evolution.

Third, future studies will investigate the integration of executive-state monitoring with digital-twin simulations, enabling controlled evaluation of intervention policies under reproducible experimental conditions.

Finally, the Cardiac Sandbox will continue to serve as the biological validation branch of the Universal Metastability Kernel, supporting future cross-domain comparisons with engineering, industrial, energy and autonomous systems while preserving a common Executive Core.

9 Validation Milestones

Table 3: Major validation milestones of the GSC-X Cardiac Sandbox.

Validation Stage	**Primary Outcome**	**Key Finding**
Event Audit V1	3216 annotated cardiac events	Established the reference distribution of normal and pathological cardiac events across the selected MIT-BIH records.
CM+ELI Executive Layer	Executive warning benchmark	Condition Metric (CM) combined with the Early Lead Indicator (ELI) enabled continuous executive-state estimation before critical cardiac events.
False Positive Audit	Alarm quality assessment	Demonstrated that a single global threshold substantially increases false-positive alarms, motivating phenotype-aware routing.
Breathing / Collapse Executive Layers	Alternative executive indicators	Multiple executive strategies were evaluated to improve the balance between detection sensitivity, lead time and alarm burden.
COLI / APB Validation	Executive refinement	Additional cardiac executive layers improved understanding of individualized cardiac dynamics and patient-specific behaviour.
Phenotype Router V3	Adaptive executive routing	Phenotype-aware routing achieved the best overall balance between precision, lead time and false-positive rate, demonstrating the benefit of adaptive executive-state selection.
Virtual Pacemaker Guard	Research architecture	Introduced a simulation framework for executive-state-guided pacing strategies based on metastability monitoring.
Universal Metastability Kernel V1.1	Cross-domain integration	Integrated Cardiac into the common Universal Metastability Kernel together with Battery, NASA CMAPSS, OpenCARP, Fusion, Industrial, Energy, Robotics and Nuclear validation domains.

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The author gratefully acknowledges the developers and maintainers of the open scientific datasets used throughout this research.

The Cardiac Sandbox validation was performed using publicly available electrocardiographic recordings from the MIT-BIH Arrhythmia Database, whose availability made this validation study possible.

The author also acknowledges the open-source scientific software ecosystem and the research communities that maintain reproducible computational tools and benchmark datasets supporting transparent scientific validation.

Artificial-intelligence assistants, including ChatGPT and Gemini, were used solely as research-support tools for technical discussion, language refinement, manuscript editing, and organizational assistance. All scientific concepts, methodology, implementation, validation, interpretation of results, and final conclusions were developed, verified, and approved by the author.

Development of the GSC-X Executive Core, the Universal Metastability Kernel, and the associated validation environments was performed independently by the author as part of the broader GSC-X research program.

Research Ethics

This work exclusively used publicly available, fully anonymized datasets.

No experiments involving human participants were performed.

No patient intervention, diagnosis or treatment was conducted.

The proposed Executive Core and Virtual Pacemaker Guard represent research algorithms intended solely for computational validation.

Data Availability

All validation datasets used in this work are publicly available.

The primary cardiac validation dataset is

MIT-BIH Arrhythmia Database

https://physionet.org/content/mitdb/

Additional validation outputs, figures and benchmark summaries are available through the GSC-X project repositories.

Reproducibility and Public Repository

Validation artifacts, benchmark summaries, technical reports and public project documentation are available at:

https://github.com/sisacek66198068/GSC-X

Interactive Demonstrator

The interactive GSC-X Cardiac Sandbox provides a live research demonstrator of the Executive Core architecture described in this work, including phenotype-aware cardiac monitoring, executive-state transitions, Metastability Reserve (MR), Metastability Risk Index (MRI), benchmark summaries and validation results.

The online demonstrator is available at:

https://cardiac.gds-gsc-x.com

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